

Course Type	Course Code	Name of Course	L	T	P	Credit
DC	MNC 300	Surface Mining	3	0	0	9

Course Objective

The objective of the course is to present an overview of surface Mining. This course initially introduces the basic concepts of surface mining and unit operations. It then exposes the students with opening of deposits using box cut. Different unit operations, allied machinery including their applicability, advantages disadvantages, method of work and productivity calculations dealt with. This course also exposes the students with different mining layouts, special mining methods and slope stability issues in surface mining.

Learning Outcomes

Upon successful completion of this course, students will:

- have a broad understanding of various unit operations and applicable machinery.
- be able to decide the method of working and applicable machinery for deployment to achieve a given production under a set of geominig condition.
- be able to estimate the production and productivity achievable with various HEMM under given conditions.
- be able to plan safe slopes in surface mines.

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1	Introduction: Surface mining – basic concepts, applicability, advantages and disadvantages. Mineral production – role of surface mining. Deposits amenable to surface mining vis-à-vis excavation characteristics. Unit operations in surface mining. Surface mining systems vis-à-vis equipment systems – classification, applicability and limitations	4	• Broad understanding of surface mines and surface mining overview vis-a-vis the mineral production. • Understanding of basic unit operations in surface mine
2	Opening up of Deposits: Box cut – objective, types, parameters and methods; factors affecting the selection of site of box-cut. Production benches – formation, parameters and factors affecting their selection.	3	• Will be able to decide the placement of initial cut for surface mining. • Will be estimate the volume of material handling for creating a box cut.
3	Preparation for Excavation: Ripper – Types, classification, applicability and limitations. Ripper – Working principle, method and cycle of operation, estimation of productivity. Concept of rippability. Estimation of number of drills required in a mine	6	• Understanding of basic functions and cutting principle of rippers. • Estimating the productivity of rippers • estimating the required number of drills for a given production.
4	Discontinuous/Cyclic Methods of Excavation and Transports: Shovel dumper operation: Applicability and limitations of electric shovels, hydraulic excavators and dumpers. Cycle time and productivity calculation for shovel and dumper. Estimation of shovel, dumper and other HEMM required for a given mine output. Method of work for sub-surface bedded and massive deposits and hilly massive deposits using shovel dumper combination. Dragline operation: Applicability and limitations, modes of operation; method of work by simple side casting Side cast diagram and calculation of reach; cycle time and productivity calculation Calculation of bucket capacity for a given handling requirement; maximum usefulness factor and its significance in dragline selection	7	• Understanding of typical machinery required in discontinuous method of mining. • Estimating the production, productivity and number of HEMM required for achieving a required production in a discontinuous system using shovel and dumper. • Basic method of work wit draglines. • Estimating the productivity and selecting the required bucket capacity.
5	Supporting Heavy Machinery: Scrappers – applicability and limitations, various types, method of work and cycle of operations, pusher dozer and push pull operation Dozers - applicability and limitations, method of work and cycle of operations, types of blade and corresponding merits and demerits Front-end-loaders - applicability and limitations, method of work and cycle of operations, minimum tipping load concept and its significance, estimation of minimum tipping load for a given front-end- loader, calculation of maximum working load bucket capacity for a given job condition.	3	• Understanding the basics of scraper, front end loaders and dozers. Their applicability and limitations. • Estimating the bucket capacity for different minerals based on the concept of minimum tipping load.
6	Continuous Methods of Excavation and Transport: Bucket Wheel Excavators: Types, principles of operation and applicability & limitations. Operational methods – Lateral/Half Block Method, Full Block Methods, and their merits and demerits. Calculation of productivity.	5	• Method of working using BWE. • Clculation of productivity of BWE. • Understanding the applicability conditions for surface miners • Basics of different types of conveyors.

	Continuous Surface Miners: Types, principles of operation and applicability & limitations. Operational methods – classification, description and their respective merits and demerits. Conveyors - operation and applicability & limitations of shiftable conveyors and High-Angle conveyors, merits and demerits of conveyor as transport system in surface mines.		
7	Semi-Continuous Methods of Excavation and Transport: Continuous excavation & partially/fully cyclic transport system – different methods and their applicability and limitations. Cyclic excavation & partially/fully continuous transport system – different methods and their applicability and limitations. In pit crushing & conveying systems – types and their applicability.	4	• Basic understanding of in-pit crushing system. • Understanding of various components of semi-continuous method of excavation. • Techno economic analysis of in-pit crushing system.
8	Mining of Developed Coal Seams and Dimensional Stones: Extraction of developed coal seams – methods, associated problems. Dimensional stones – types, occurrences and uses. Methods vis-à-vis equipment used for extraction of primary blocks in granite marble quarries	5	• Precautions in working developed coal seams using surface mining method. • Basic unit operations in dimensional stone quarries.
9	Slopes in Surface Mines: Types of mine slope – highwall and waste dumps, common modes of slope failure. Factors influencing stability stability of slopes. Slope stability assesment techniques. Types and formation methods of waste dumps. Protection, stabilization and monitoring of slopes	5	• Types and formation of slopes • assessment of stability of highwall and dumps
TOTAL		42	

Text Books

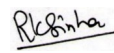
1. Surface mining [edited by] Eugene P. Pfeider. American Institute of Mining, Metallurgical, and Petroleum Engineers, 1968, New York
2. Surface mining by Bruce A Kennedy; Society for Mining, Metallurgy, and Exploration (U.S.), Publisher: Littleton, Colo. : Society for Mining, Metallurgy, and Exploration, 1990.
3. Surface Mining Equipment Hardcover –1982 by James W. Martin
4. Surface Mining by G B Mishra

Reference:

1. SME Handbook by H L Hartman

Evaluation:

1. Marks Distribution
 - (a) Mid Sem : 32%
 - (b) End Sem : 48%
 - (c) Assignments : 20%
2. There is no weightage for attendance in marks distribution.



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